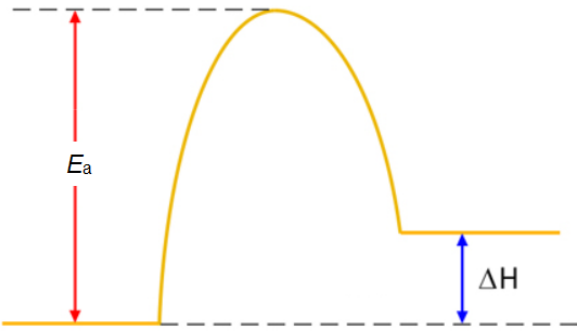
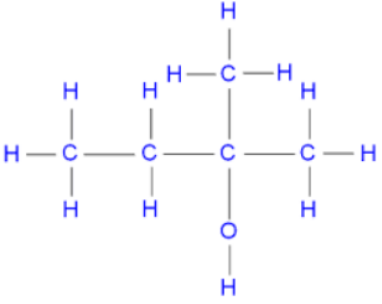


SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				2-methylpentane	1			1		
2				 award (1) for profile shape (products above reactants) award (1) for E_a and ΔH labelled – both needed	2			2		
3				molecular formula $C_6H_{10}O_2$ (1) empirical formula C_3H_5O (1)		2		2		
4	(a)			there are two hydrogen atoms on one of the carbons		1		1		
	(b)			$\begin{array}{cc} \text{H} & \text{Cl} \\ & \\ -\text{C} & -\text{C}- \\ & \\ \text{H} & \text{H} \end{array}$		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5				$\text{CH}_3\text{CHBrCH}_3$ accept any unambiguous structure for 2-bromopropane	1			1		
6				 accept any unambiguous structure for 2-methylbutan-2-ol		1		1		
7				nickel	1			1		
				Section A total	5	5	0	10	0	0